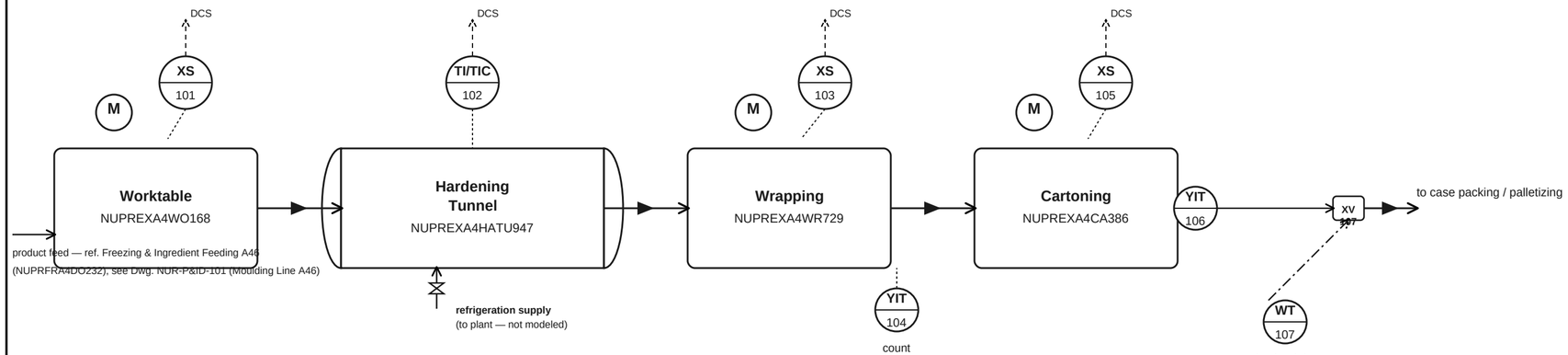
P&ID (PRELIMINARY) — NUREMBERG SITE — EXTRUSION LINE A46 — ICE CREAM FACTORY DATA

Cognite Data Fusion Core Data Model | ISA-5.1 instrument tagging | For L100 practitioner training use only — not for construction



Legend

Solid line + ►	Process piping / product flow (arrowheads = assumed direction)
Bowtie (on a line)	Manual isolation valve
Short dashed line	Instrument signal to PLC / DCS
Dash-dot line	Safety interlock / control logic
Circle (M)	Electric motor / servo drive
TI / TIC	Temperature indicator / temperature indicating controller
XS / YI	Motor state switch / status indicator
YIT / YC	Event or pulse counter (count variable)
WT	Checkweigher (quality variable)
XV / YV	Solenoid reject valve / pneumatic pusher

Notes / open items before this drawing is construction-ready

Process order (Worktable -> Hardening Tunnel -> Wrapping -> Cartoning) is inferred from standard extrusion-line practice — the source data gives no explicit sequence, only a flat unit list.

This line has only 4 units in the source data — no separate conveyor/transfer asset exists (unlike Moulding's Main Drive or Filling's Transfer System). Connectors are drawn as plain piping but likely represent mechanical transfer in reality.

Hardening Tunnel's temperature control (TI/TIC-102) and refrigeration supply tie-in are inferred — no piping diameters or setpoints exist in the source data (contrast with the Continuous Freezer's real diameters on Dwg. NUR-P&ID-000).

Worktable's product feed references the Freezing & Ingredient Feeding A46 cell rather than redrawing it — see Dwg. NUR-P&ID-101 (Moulding Line A46) for that equipment's detail.

This line has no vessels of its own — nothing here to vent, drain, or sample.

Wrapping here is the same unit type/spec as on Moulding and Filling Lines A46. Cartoning has no machine spec in the source data (unlike the other two lines' Cartoning, model SA-380CDF).

planned_status (Core Data Model time series) is a software/planning variable only — intentionally not drawn on this P&ID.

Instrument loop numbers (101-107) are illustrative for this training exercise, not an assigned plant numbering series.

Source: Ice Cream Factory data, Nuremberg site, Extrusion Line A46, modeled via the Core Data Model (ProductionLine / Unit / Equipment).

Drawing No.	NUR-P&ID-102
Client / project	Ice Cream Factory — L100 practitioner training
Site	Nuremberg
Unit	Extrusion Line A46
Drawing type	P&ID — preliminary (ISA-5.1)
Data source	Core Data Model (ProductionLine, Unit, Equipment)
Status	For training use — not for construction
Rev.	0 — draft